Quantum Computation and Information

Homework Set 5

1. Recall that there are 8 stabilizers for Shor's 9-qubit error correcting codes, and Shor's codes consist of three 3-qubit clusters.

   (a). Give a set of 8 stabilizers and show that they are commutative.

   (b). Suppose that, without loss of generality, there is a bit-flip error in the first physical qubit. What is the corresponding error operator? Which stabilizers are anti-commutative with this error operator?

   (c). Suppose that, there is a phase-flip error in one of the qubit of the first cluster. What is the corresponding error operation? Which stabilizers are anti-commutative with this error operator?

   (d). Show that the distance of Shor's codes is 3.

2. Consider the surface code on a $L \times L$ lattice where we identify the left-most and right-most qubits, as well as the top-most and bottom-most qubits (we get a torus). What is the $[n, k, d]$ parameters of this surface code?

3. Consider the stabilizer group $\mathcal{S}$ that contains all the stabilizers S_i and their compositions. An operator L is an logical operator if $[L, S_i] = 0$ for all S_i. Show that a logical operator maps the codespace to itself.

4. **Half-way teleportation.** Alice and Bob share an unknown state $|\psi\rangle$ (Alice holds the first qubit and Bob the second). They want to perform the unitary operator

$$\Gamma^2(e^{i\phi}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\phi} \end{pmatrix}$$

   This is easy to do if Alice and Bob also share two EPR pairs and have access to a classical communication channel: Alice teleports her qubit to Bob; he applies the gate and teleports the qubit back. How do they achieve their goal using only one EPR pair?

   Hint: Alice does some partial measurement on her qubit and her part of the EPR pair, and tells the result to Bob. (Here "partial measurement" means the projection onto a pair of orthogonal two-dimensional subspaces of $\mathbb{C}^2 \otimes \mathbb{C}^2$; quantum coherence within each subspace is preserved.) Having received one classical bit from Alice, Bob creates a "copy" of Alice's qubit relative to the standard basis, $|x\rangle \to |xx\rangle$. The required gate can now be realized in a straightforward way. The last part is to erase the extra copy of x. To achieve that, Bob does another partial measurement and communicates the result to Alice, who applies a suitable gate to her qubit. Thus, the problem splits into tasks: (i) create a remote copy (relative to the standard basis) using one EPR pair and one bit of communication; (ii) erase such a copy while maintaining quantum coherence, which involves sending one bit in the opposite direction.